Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/13/2024
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Leveraging Spaceflight Genomic Data to Uncover Developmental and Cell Type Specific Gene Regulatory Networks in Plants Responding to Gravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chin, Sabrina Ph.D. (University of Wisconsin-Madison)		
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Task Description:	As NASA pursues its goal of sending human colonies to Mars and beyond, there is a pressing need to better understand how the spaceflight environment influences fundamental biological processes. Understanding how plants respond to gravity on Earth and the microgravity environment in space is of particular interest because of the important role that plants will play in advanced bioregenerative life support systems during long duration space missions. In recent years, numerous transcriptomic datasets have shown that hundreds of genes change expression when plants are grown in space. However, it has been challenging to tease apart which genes change due to microgravity from those that change as a result of general plant stress responses. This difficulty stems from the fact that plant spaceflight genomics datasets have been derived almost exclusively from whole seedlings and plant organs, which could potentially mask or dilute gene expression changes that are occurring within the distinct cell-types that are responsible for gravity sensing (e.g., root columella and stem endodermal cells). Thus, the major goal of this NASA-ground-based proposal is to begin to take steps to develop methodology that will enable space genomic studies on homogenous cell populations. Here, we propose to use laser capture microdissection (LCM) to conduct RNA-Sequencing (RNA-Seq) on gravity sensing columella cells to identify gene regulatory networks that govern early responses of plants to gravity. We hypothesize that gene profile signatures from a homogenous population of columella cells will more clearly reflect transcriptomic changes that are directly linked to microgravity. The RNA-Seq data that we obtain from our analysis of root columella cells will be compared to columella-specific gene modules we identified through our Weighted Gene Co-expression Network Analysis (WGCNA) of Arabidopsis seedlings grown for 6 and 11 days on the International Space Station (ISS) (specific aim 2). In parallel, we will refine LCM methods with the goal of implementing them for future plant spaceflight genomics (specific aim 1). The latter aim will involve ground-based experiments in which specific steps in the LCM methods will be modified to mirror in-orbit operations. Upon successful completion of this proposal, we expect to generate new tools for conducting genomics in homogenous plant cell populations that will enable us to better differentiate microgravity effects from microgravity independent effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research seeks to fill technological gaps that will enable a better understanding of how plants develop in the microgravity environment of spaceflight. The focus of this research is to develop methods that will allow us to learn more about how expression of genes within specific root cell types change under microgravity or stress-induced spaceflight. The plant organ of interest in this proposal is the root. Because roots are crucial for uptake of water and nutrients to support optimal plant growth, results from this proposal will benefit agriculture on Earth. We expect that genes, which show cell-type expression changes in response to gravity could be potential targets for pre-breeding efforts toward developing crop cultivars that are resilient to environmental stress imposed by marginal lands on Earth and confined growth environments during spaceflight.
	SPECIFIC RESEARCH AIMS Aim 1: Refine and develop laser capture dissection (LCM) methodology in plant roots for potential implementation in future spaceflight studies. Aim 2: Characterize differential gene expression within gravity sensing columella cells during root gravitropism. Aim 3: Characterize the function of candidate gene(s) that demonstrated differential expression in maize root cells during gravity stimulation. Aim 4: Perform a parallel study of maize orthologous gene(s) with likely function in gravitropism in Arabidopsis thaliana. RESEARCH ACCOMPLISHMENTS This grant is a continuation of NASA grant 80NSSC19K0129, which was transferred from the Noble Research Institute to the University of Wisconsin in May 2022 (UW; current grant 80NSSC22K0024). Research accomplishments at the Noble Research Institute and since the grant transfer to the University of Wisconsin are outlined below. Specific Aims 1 & 2: Refine and develop laser capture microdissection (LCM) methodology in plant roots for potential implementation in future spaceflight studies and utilize this approach to characterize differential gene expression within gravity sensing columella cells during root gravitropism. Root caps consist of longitudinally arranged cells which include the meristem, columella, and lateral root cap cells. The columella cells contain amyloplasts that act as statoliths for gravity perception in the roots (Ijima et al., 2008). Root caps can be collected non-destructively by manually dissecting them with a scalpel. Using this approach proved to be an easier and much more high throughput alternative to LCM, making it compatible with isolating sufficient samples enriched for gravity sensing cells in the root cap to use for RNA sequencing (RNA-Seq)-based transcriptomic analyses. Maize roots are relatively large, with root caps harvestable by cutting ~3 mm of the tip of the root. This approach yielded samples suitable for extraction of high-quality RNA that was used for bulk RNA-Seq to analyze for genes that were differentially expressed in the root tip when plants were gravi-stimulated by rotating by 90 degrees, versus plants maintained vertically. Analysis of the transcriptome response confirmed that the gravistimulation led to different gene expression patterns with a distinct separation of vertical control and gravity simulated data points. These gravity-related changes were also seen to change over time. Root tip cells displayed the most transcriptional changes at 30 minutes post gravity stimulation, with 47 significantly differentially expressed genes (DEGs). A second peak in DEGs was seen at 2 hours, with 29 differentially expressed genes. The DEGs at the peak in transcriptional reprogramming at 30 min were analyzed for their functional groupings by Gene Ontology (GO) enrichment analysis and were related to metabolism, cell cycle, energy, and the cell wall. The lack of detectable changes in transcription at 5- and 10-minutes post gravistimulation may indicate the lag between events related to gravity perception and transcriptional activation. Nonetheless, studies have shown that the gravity signal is transduced by root cap cells within 15 minutes of gravity stimulation as they exhibit dynamic changes in signaling elements, such as Ca2+ (Plieth and Trewavas, 2002) and microtubule dynamics (Blancaflor and Hassenstein, 1997). Indeed, lines expressing the GCaMP Green Fluorescent Protein-based Ca2+ sensor in a root tissue specific manner, including the gravity sensing cells of the root cap, have been generated as part of this work to allow confirmation of

Task Progress:

these rapid signaling events (Krogman et al., 2020).

The observation that rapid signaling occurs within this 15 min timeframe where no significant transcriptional response was observed may indicate that the root cells preferentially utilize a rapid, non-transcriptional signaling pathway for these earlier, perception-related time points. The large number of DEGs at 30 min would then reflect a wave of transcriptional response to these early signaling events. The lack of DEGs at 1 hour post gravity stimulation may represent a resetting point of earlier gravity signal perception to allow for a shift from perception-linked events to the regulation differential growth observed at 2 hours. The gene ontologies at this 2 hour timepoint were cell cycle, transcription, metabolism, energy, and the cytoskeleton.

Since 30 mins and 2 hours exhibited the most DEGs, these timepoints were the focus of analysis using two linear modeling packages for analyzing RNA-Seq data, DESeq2 (Love et al., 2014) and limma-voom (Law et al., 2014) implemented in the iDEP 2.0 software (Ge et al., 2018). Gene candidates were further filtered based on gene annotations in maize and available orthologue information in Arabidopsis thaliana (A. thaliana), alongside the availability of knockout mutants in both species. This approach yielded 7 high priority candidates for further study.

Specific Aims 3 & 4: Characterize the function of candidate gene(s) that demonstrated differential expression in maize root cap cells during gravity stimulation and perform a parallel study of maize orthologous gene(s) with likely function in gravitropism in Arabidopsis thaliana.

It was possible to isolate homozygous maize mutants in 3 of the 7 candidate genes identified from aims 1 and 2: TIR2, NPI270, and OSCA5. These mutants were then phenotyped for root gravitropic response in comparison to their wild-type parent (the maize W22 inbred line). The TIR2 mutant showed a significant reduction in gravitropic root bending between 1 and 2.5 hours after the root was rotated by 90°.

Orthologous genes to the maize genes outlined above were also identified in Arabidopsis thaliana and homozygous knockout mutant lines isolated for gravitropism assay. The A. thaliana orthologs of maize TIR2 are the AUXIN F-BOX (AFB) 2 and 3 genes. These belong to the family of TIR1/AFB auxin receptors that localize in the root (Prigge et al., 2020). Genetic lesions of AFB2 and AFB3 in single and double mutants also resulted in reduced root gravitropic response, similar to the lowered gravitropic response seen in tir2.

As AFBs are intimately linked to auxin response, these Arabidopsis knockout mutants were crossed with plants expressing the fluorescent auxin reporter DII-Venus. Analysis of DII-Venus signal then allowed analysis of whether reduced gravitropic response is likely due to changes in auxin response, in particular altered auxin asymmetry between the upper and lower flank of the root. This bioreporter is targeted to the nucleus where it fluoresces. However, as auxin levels rise, the reporter is degraded. Analysis of the wild-type DII-Venus response showed that, upon reorienting the root to the horizontal, the expected asymmetry in reporter developed over a 3 h timeframe, with the lower side of the root showing reduced signal, i.e., reflecting the predicted increased auxin levels on the lower side of the root that are known to drive gravitropic growth. However, the development of this asymmetry appeared to be lost between 1 and 2 hours in afb2-3 and afb3-5 mutants. This observation suggests that the lesion in gravitropic response in afb2 and afb3 lines might be linked to a feedback regulation of these AFBs on the auxin transport and redistribution required for differential growth during root bending.

IN SUMMARY, the major findings from this research have been: (1) transcriptomic changes elicited by gravity stimulation of the maize root occur in 2 phases. Phase 1 occurs at ~30 min after initial perception events and phase 2 occurs after ~2 h and is likely related to the modulation of cell growth. There is a time between these 2 phases when gravistimulation-triggered gene expression falls to zero, possibly reflecting a transcriptional reprogramming away from perception events towards gravitropic growth control. (2) TIR2/AFB2/AFB3 are involved as regulators of root gravitropic growth. (3) AFB2/AFB3 may be operating by modulating the auxin redistribution that is triggered by gravity stimulation, likely in a feedback loop on auxin production or transport.

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Articles in Peer-reviewed Journals	Bakshi A, Swanson SJ, Gilroy S. "A touchy subject: Ca2+ signaling during leaf movements in Mimosa." Cell Calcium. 2023 Mar;110:102695. https://doi.org/10.1016/j.ceca.2023.102695 ; PMID: 36669253 , Mar-2023
Books/Book Chapters	Chin S, Johns S, Gilroy S. "Calcium in root growth and development " in "Plant Roots: The Hidden Half, Fifth Edition (5th ed.)." Ed. A Eshel, T Beeckman. Boca Raton, FL, CRC Press, 2024. p. 289-304. https://doi.org/10.1201/b23126 , Aug-2024